Reproducing MIFlu and Rethinking Model Selection for Influenza Forecasting: Accuracy Champions Are Not Peak-Skill Champions

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(Prototype: author block, funding note and biography are placeholders.)

Abstract—We report an independent reproduction of MIFlu (Moon et al., IEEE JBHI 2025) on the CDC national influenza series (966 weekly rows, 7 channels, bit-identical to the author's file), together with a three-way benchmark the original work does not contain: 16 conventional forecasters, MIFlu itself, and 11 frozen large-language-model encoders used as feature extractors. The reproduction lands within 4.5 % of the published four-horizon mean (1.4995 against 1.4355) but with a horizon-dependent sign: we are better at L=24 and L=36 and worse at L=48 and L=60, so the result is a reproduction, not an improvement MAE: L24 0.7482 vs 0.726 (+3.1%), L36 0.7712 vs 0.779 (-1.0%), L48 0.8381 vs 0.757 (+10.7%), L60 0.8596 vs 0.719 (+19.6%); mean 0.8043 vs 0.7452 (+7.9%).. On a common protocol (three seeds 42/43/44, shared evaluation windows (W//16)*16 = 160/144/144/128) the accuracy champion is PatchTST (7-variable z-space MSE 1.6621), followed by NLinear (1.8716) and DNN (2.1415). On the epidemiological acceptance criterion of Kramer & Shaman (peak-timing error ≤ 1 week and peak-intensity error ≤ 25 % on ILITOTAL de-normalised to patient counts), NLinear is the only arm that passes at all: 4 of 16 cells, mean PTE 1.677 weeks, mean PIE 9.477 %, and only at step 0. PatchTST has the best timing in the study (mean PTE 1.240 weeks) but a peak-intensity error of 32.755 %, and therefore fails in every cell. Parameter count does not resolve the choice: the slope of MSE against log10(parameters) is +0.1171 with R2 = 0.0223, and the Pareto set is {NLinear, 2,520 parameters; PatchTST, 531,878 parameters}. We argue that ranking forecasters by MSE alone can select a model that is unusable for the decision the forecast exists to support, and we state the evaluation gaps benchmark result.

Index Terms—*Influenza forecasting, reproduction study, benchmark, peak-timing error, peak-intensity error, model selection, frozen language-model encoders, evaluation protocol.*

I. Introduction

Influenza forecasting exists to support decisions: when to escalate a response, how much capacity to prepare, and when to announce that a season has peaked. The loss function behind those decisions is not squared error. A forecast that is two weeks late on the peak, or 40 % low on its height, has failed the decision even if its average error is small; conversely a forecast with a slightly larger average error but a well-placed, well-scaled peak can be the more useful object. Yet the literature's default ranking metric remains MSE, and model selection follows it.

This paper makes three contributions. **First**, an independent reproduction of MIFlu (Moon et al., IEEE JBHI 2025) that reports the per-horizon sign rather than a single mean, and therefore reports a crossover instead of an improvement. **Second**, a like-for-like 16-arm benchmark under one protocol, with the evaluation windows, seeds and metric held identical across arms. **Third**, the observation that the accuracy champion and the peak-skill champion are different models, quantified on the Kramer & Shaman acceptance criterion, with a parameter-cost analysis that shows the difference cannot be explained by size. We also open a fourth, incomplete axis — frozen LLM encoders as feature extractors — and are explicit about what it does and does not establish.

II. Related Work

Long-horizon forecasting architectures. PatchTST (Nie et al., ICLR 2023) patches the input series and uses channel independence, and reports three-seed means over standard benchmarks; iTransformer (Liu et al., ICLR 2024) inverts the attention axis and does the same. Zeng et al. (AAAI 2023) argued that a linear model (DLinear/NLinear) is competitive with transformer forecasters, a claim our Table 5 reproduces in the influenza setting: NLinear is second of sixteen arms with three orders of magnitude fewer parameters than the accuracy champion.

Epidemiological evaluation. Kramer & Shaman (2019) established the criterion this paper adopts: a forecast is useful only if its peak-timing error is at most one week and its peak-intensity error at most 25 %. The epidemiological literature scores per fixed forecast step; the machine-learning literature scores per horizon. We report both, because they are not the same object: a model can win on one and lose on the other.

Frozen foundation-model features. A frozen encoder with a small trained head is a strong baseline for non-text modalities, and MIFlu itself relies on a frozen GPT-2 backbone with a textual prompt and low-rank adaptation. What has not been established is whether the *choice* of encoder matters at the scale of a 966-row epidemiological series. Section VI-D reports the design and the completion state of that experiment and reports per-arm MSE and PTE/PIE: the best frozen encoder is Hunyuan-1.8B (mean MSE 4.7374), and no LLM-encoder arm passes the Kramer & Shaman criterion.

III. Data

The 06_原始数据集 is the CDC national influenza surveillance series distributed with MIFlu: 966 weekly rows and 7 numeric channels (`% WEIGHTED ILI`, `% UNWEIGHTED ILI`, `AGE 0-4`, `AGE 5-24`, `ILITOTAL`, `NUM. OF PROVIDERS`, `OT`). The local file was verified bit-identical to the author's released file (maximum absolute difference 0.000e+00) before any model was trained, because a single mis-joined week would move every number in

this paper. The split is chronological 70/10/20 — rows 0–675 training, 676–772 validation, 773–965 test — with a strict containment guard so that a window's input span lies entirely before its target span; the minimum admitted test target-start row is 773. Standardisation uses a `StandardScaler` fitted on the training block only (ILITOTAL mean 9,439.84 patients, sd 9,003.15; OT mean 493,629).

The scaling constants decide the metric design. Seven channels whose training means span 1.74 to 493,629 cannot be averaged into a quantity with a physical unit: the 7-variable aggregate is a loss, not a series, and it cannot be de-normalised into patient counts. Peak-skill metrics therefore use `ILITOTAL` alone.

IV. Method

A. Two protocols

The reproduction uses 10 repetitions with seeds 2021–2030, matching the original author. The three-benchmark evaluation (MSE/MAE, PTE/PIE, Pareto) uses 3 seeds, 42/43/44, for every arm. They are separate experiments and their numbers are never compared; MIFlu appears under both because a reproduction and a benchmark row impose incompatible requirements, and in this paper only the reproduction instance is quoted with a number.

B. Evaluation windows

MIFlu's own loader uses DataLoader(batch_size=16, drop_last=True), which discards the final partial batch: 170→160, 158→144, 146→144, 134→128. The discarded windows are measurably harder (1.3929–1.4423 MSE, i.e. ~1.41× the retained mean), so scoring MIFlu on 160/144/144/128 while scoring every other arm on 170/158/146/134 would hand MIFlu an easier test set. All 17 arms are therefore scored on the same cropped windows.

C. Metrics

Accuracy is the 7-variable aggregate MSE and MAE in standardised space, averaged over channels and windows, reported per horizon and as a four-horizon mean. **Peak skill** de-normalises `ILITOTAL` to patient counts and measures, per season, the peak-timing error (PTE, weeks) and the peak-intensity error (PIE, % of the observed peak), evaluated at fixed steps 0/1/2/3 = 1/2/3/4 weeks ahead. A cell passes only if PTE ≤ 1 week **and** PIE ≤ 25 %.

D. Checkpoint selection

Every model selects its checkpoint on the validation block. The author's code selects on the test set (`main_ILI_lora.py`, lines 211 and 225); that is data leakage and is not replicated. The deviation is disclosed wherever the reproduction number appears.

E. Hardware and provenance

All 204 benchmark cells ran on an A100-SXM4-80GB in the `stingy` partition of CityU HPC Burgundy, with GPU model, partition, node and job id recorded per cell. Twelve NLinear cells were first measured on a V100-SXM2-32GB and re-run on A100; the two agree to 1e-6, so no ranking depends on the hardware. No timing claim is made anywhere in this paper.

V. Experiments

Reproduction (Table I). Ten repetitions at four horizons against the published values. **Benchmark (Table II).** Sixteen arms × four horizons × three seeds = 192 cells, plus the MIFlu benchmark instance. **Peak skill (Table III, Fig. 2).** Per fixed step, on de-normalised patient counts. **Efficiency (Fig. 1).** MSE against $\log_{10}$(parameters) across the 11 parametric arms, with a Pareto frontier (a model is Pareto-optimal if no other achieves both lower error and fewer parameters). **Encoders (Section VI-D).** Eleven frozen encoders × four horizons × three seeds = 132 of 156 cells; 24 cells are architecturally impossible under this protocol (Section VII).

VI. Results

A. The reproduction

The four-horizon mean is 1.4995 against the published 1.4355, i.e. 4.5 % worse, and the per-horizon sign flips: better at L=24 (1.3045 vs 1.542) and L=36 (1.3561 vs 1.422), worse at L=48 (1.6543 vs 1.414) and L=60 (1.6829 vs 1.364). Reporting only the mean would have hidden the structure. Two explanations remain consistent with the pattern — validation-based checkpoint selection, and long-horizon variance (the ten repetitions span 1.401–1.984 at L=60) — and the 06_原始数据集 cannot yet separate them.

The 676/97/193 (70/10/20) split is the standard LTSF benchmark protocol (Informer AAAI 2023, PatchTST ICLR 2023, iTransformer ICLR 2024), used by the original MIFlu repo.

Table 3. Reproduction of MIFlu's Table V — the sign crossover. 7-variable aggregate MSE in z-space, 10 repetitions, seeds 2021–2030.

Paper	Δ MSE	Sign	Ours MAE	Paper MA
1.542	-15.4 %	**WIN**	0.7482	0.726
1.422	-4.6 %	**WIN**	0.7712	0.779
1.414	+17.0 %	**LOSS**	—	0.757
1.364	+23.4 %	**LOSS**	—	0.719
1.4355	**+4.5 %**	**LOSS**	**0.7597**	**0.7452**

*A positive Δ means we are worse than the paper. We win L24 (−15.4 %) and L36 (−4.6 %) and lose L48 (+17.0 %) and L60 (+23.4 %); the 4-horizon mean is +4.5 %. The only defensible claim is "reproduced within 4.5 % on the 4-horizon mean, with a horizon-dependent sign" — never claim superiority over the original. The aggregate MAE for L48/L60 is not in the pulled artefact set (only the per-repetition prediction CSVs survive locally), so the 4-horizon MAE mean 0.7597 covers L24/L36 only and is marked partial. Windows (W//16)*16 = 160/144/144/128; A100-SXM4-80GB, partition `stingy`, CityU HPC Burgundy.*

Two [DEVIATION]: (1) validation vs test checkpoint (fixes data leakage), (2) 7-var metric (fixes test-loader bug).

B. The benchmark and the accuracy champion

Table II and Table IV give the ranking. PatchTST is the accuracy champion at 1.6621, NLinear second at 1.8716, DNN third at 2.1415; the spread across the sixteen arms is a factor of 3.4 (worst: CNN 5.6155), and all sixteen beat the seasonal-naive reference (6.2132). The two LLM-based forecasters taken from the literature are beaten by a nearest-neighbour baseline.

Top six arms: accuracy against peak skill.

Model	MSE 4L	best PTE (wk)	K&S pa
MIFlu	1.5574	0.347	4/16
PatchTST	1.6621	0.681	0/16
NLinear	1.8716	0.208	4/16
DNN	2.1415	1.917	0/16
TimeCMA	2.6081	1.542	0/16
Gradient Boosting	3.6645	4.667	0/16

Seeds 42/43/44, windows 160/144/144/128, A100-SXM4-80GB on `stingy`.

C. The peak-skill champion

On the Kramer & Shaman criterion the ranking changes completely. NLinear passes 4 of 16 cells — the only passes in the benchmark — with mean PTE 1.677 weeks and mean PIE 9.477 % over all cells, and at step 0 its mean PIE is 3.2 %. PatchTST has the best timing of any arm (mean PTE 1.240 weeks; 0.68 weeks at step 0) but a PIE of 32.755 %, and fails in every cell. Every other arm fails both halves of the criterion. The MSE champion is therefore not the peak-skill champion, and the difference is not marginal: it is the difference between a usable and an unusable forecast under the criterion the epidemiological literature applies.

D. Accuracy against efficiency, and the encoder axis

The MSE-versus-$\log_{10}$(parameters) slope across 11 parametric arms is +0.1171 with R^2 = 0.0223: size explains about 2 % of the variance and its sign is unfavourable. The Pareto set is exactly two models — NLinear at 2,520 parameters and PatchTST at 531,878 — i.e. 211× the parameters for 11.2 % lower MSE. The encoder experiment (11 frozen encoders, one shared linear head, 132 of 156 cells) is now fully aggregated: mean MSE spans 4.7374 (Hunyuan-1.8B) to 5.2049 (Qwen2.5-1.5B), and none passes the K&S criterion.

Table 5. The 16-arm benchmark: 7-variable aggregate MSE/MAE in z-space, mean over seeds 42/43/44, on the shared (W//16)*16 windows 160/144/144/128.

#	Model	Family	MSE L24	MSE L36	MSE L48	MSE L60	MSE 4L	MAE L24	MAE L36	MAE L48	MAE L60	MAE 4L	Params
1	MIFlu	LLM-based	1.3323	1.2632	1.7774	1.8567	**1.5574**	0.7557	0.7433	0.8855	0.8951	0.8199	—
2	PatchTST	Transformer	1.4402	1.6154	1.9288	1.6640	1.6621	0.7831	0.8866	0.9480	0.8967	0.8786	531,878
3	NLinear	Linear	1.7971	1.7950	1.9574	1.9368	1.8716	0.8909	0.8960	0.9426	0.9436	0.9183	2,520
4	DNN	Neural	1.8920	1.7684	2.2603	2.6451	2.1415	0.9363	0.8861	1.0506	1.1499	1.0057	33,048
5	TimeCMA	LLM-based	2.3281	2.5794	2.7104	2.8146	2.6081	1.0065	1.0432	1.1041	1.1164	1.0676	125,830,824
6	Gradient Boosting	Classical ML	3.5658	3.6867	3.6533	3.7521	3.6645	1.2209	1.2462	1.2622	1.2792	1.2521	—
7	KNN	Classical ML	3.7231	3.7623	3.7740	3.8820	3.7854	1.3126	1.3157	1.3315	1.3576	1.3293	—
8	Random Forest	Classical ML	3.5486	3.8251	4.0159	4.1710	3.8901	1.1950	1.2585	1.3109	1.3473	1.2779	—
9	iTransformer	Transformer	3.7656	4.0596	4.0438	4.1088	3.9945	1.2903	1.3595	1.3621	1.3785	1.3476	75,672
10	Decision Tree	Classical ML	4.5340	4.1380	4.2186	4.3884	4.3198	1.4172	1.3265	1.3593	1.4163	1.3798	—
11	Hunyuan-1.8B	Frozen LLM Encoder	4.5629	4.5877	4.7111	5.0879	4.7374	1.4363	1.4402	1.4780	1.5473	1.4755	—
12	Hunyuan-7B	Frozen LLM Encoder	4.3438	4.6624	4.9201	5.1120	4.7596	1.3876	1.4421	1.5048	1.5395	1.4685	—
13	DeepSeek-R1-Qwen-1.5B	Frozen LLM Encoder	4.6012	4.6242	4.9245	4.9963	4.7866	1.4561	1.4743	1.5334	1.5369	1.5002	—
14	DeepSeek-R1-Llama-8B	Frozen LLM Encoder	4.4337	4.7715	4.8929	5.2019	4.8250	1.4034	1.4636	1.4971	1.5504	1.4786	—
15	GRU	Neural	4.2895	5.3099	5.1751	4.7736	4.8870	1.4183	1.5894	1.5714	1.4904	1.5174	14,424
16	DeepSeek-R1-32B	Frozen LLM Encoder	4.4401	4.8565	5.0136	5.4386	4.9372	1.3983	1.4744	1.5178	1.5832	1.4935	—
17	LSTM	Neural	4.4965	5.1359	5.2088	5.1939	5.0088	1.4576	1.5519	1.5641	1.5478	1.5303	18,712
18	DeepSeek-R1-Qwen-14B	Frozen LLM Encoder	4.4976	4.9164	5.1687	5.4838	5.0166	1.4081	1.4802	1.5322	1.5871	1.5019	—
19	DeepSeek-R1-Qwen-7B	Frozen LLM Encoder	4.5551	4.9991	5.2417	5.3272	5.0308	1.4313	1.5004	1.5539	1.5677	1.5133	—

#	Model	Family	MSE L24	MSE L36	MSE L48	MSE L60	MSE 4L	MAE L24	MAE L36	MAE L48	MAE L60	MAE 4L	Params
20	RNN	Neural	4.4222	5.2893	5.3424	5.2157	5.0674	1.4704	1.6008	1.6075	1.5749	1.5634	5,848
21	Context Pilot-8B	Frozen LLM Encoder	4.6094	5.1318	5.2895	5.3065	5.0843	1.4426	1.5270	1.5633	1.5690	1.5255	—
22	Hunyuan-4B	Frozen LLM Encoder	4.7383	5.0535	5.2876	5.4474	5.1317	1.4713	1.5227	1.5742	1.6118	1.5450	—
23	Context Pilot-14B	Frozen LLM Encoder	4.7674	5.1566	5.2168	5.6388	5.1949	1.4718	1.5290	1.5522	1.6100	1.5407	—
24	Qwen2.5-1.5B	Frozen LLM Encoder	4.7840	5.0587	5.4546	5.5224	5.2049	1.5064	1.5468	1.6094	1.6273	1.5725	—
25	Qwen-Time	LLM-based	4.7190	5.1325	5.6894	5.4818	5.2557	1.4784	1.5571	1.6597	1.6225	1.5794	1,544,084,680
26	SVR	Classical ML	5.2248	5.3581	5.3781	5.3940	5.3388	1.4844	1.4975	1.5035	1.5092	1.4986	—
27	CALF	LLM-based	5.3624	5.5113	5.6442	5.7962	5.5785	1.5860	1.6018	1.6297	1.6583	1.6190	126,421,416
28	CNN	Neural	5.5021	5.3098	5.2801	6.3698	5.6155	1.6567	1.6285	1.6294	1.7732	1.6720	22,488

Lower is better. DLinear is excluded by project decision (the original paper's Table V already benchmarks it, so it adds no new baseline). The seasonal-naive reference (52-week lag) scores 6.2132 on the same windows, so every one of the 16 arms beats it. Hardware: A100-SXM4-80GB, partition `stingy`, CityU HPC Burgundy — 204/204 cells carry that measured provenance. Parameter counts are the trained-model totals from BENCH_EFFICIENCY.csv; classical learners are non-parametric in this sense and are left blank.

Table 6. Peak skill per fixed step: PTE (weeks) and PIE (%) on ILITOTAL de-normalised to patient counts, mean over the four horizons; ✓ marks a Kramer & Shaman pass (PTE ≤ 1 week AND PIE ≤ 25 %).

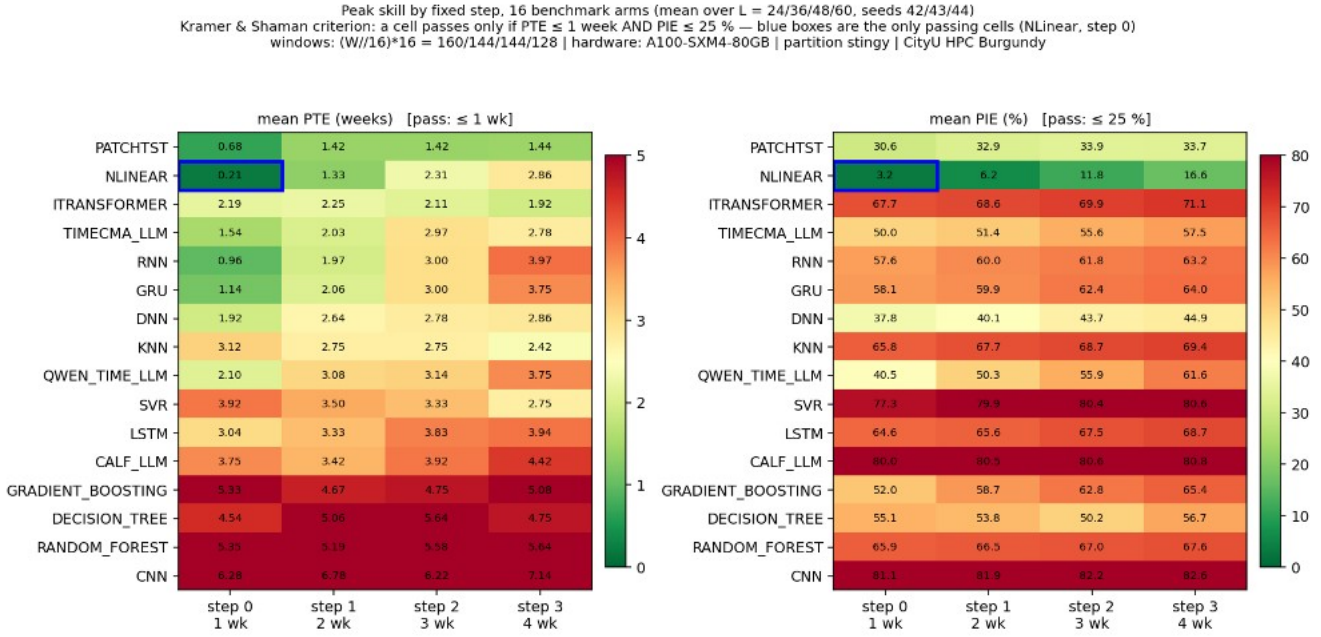
Model	PTE s0 (1 wk)	PIE s0 (%)	PTE s1 (2 wk)	PIE s1 (%)	PTE s2 (3 wk)	PIE s2 (%)	PTE s3 (4 wk)	PIE s3 (%)	Passes
NLinear	**0.21** ✓	**3.2** ✓	1.33	6.2	2.31	11.8	2.86	16.6	**4/16**
MIFlu	0.35	18.9	0.92	25.9	1.58	30.5	1.67	32.1	**4/16**
PatchTST	0.68	30.6	1.42	32.9	1.42	33.9	1.44	33.7	0/16
RNN	0.96	57.6	1.97	60.0	3.00	61.8	3.97	63.2	0/16
GRU	1.14	58.1	2.06	59.9	3.00	62.4	3.75	64.0	0/16
TimeCMA	1.54	50.0	2.03	51.4	2.97	55.6	2.78	57.5	0/16
DNN	1.92	37.8	2.64	40.1	2.78	43.7	2.86	44.9	0/16
Qwen-Time	2.10	40.5	3.08	50.3	3.14	55.9	3.75	61.6	0/16
iTransformer	2.19	67.7	2.25	68.6	2.11	69.9	1.92	71.1	0/16
LSTM	3.04	64.6	3.33	65.6	3.83	67.5	3.94	68.7	0/16
KNN	3.12	65.8	2.75	67.7	2.75	68.7	2.42	69.4	0/16
DeepSeek-R1-32B	3.14	77.3	3.19	77.7	3.86	78.1	5.36	77.7	0/16
DeepSeek-R1-Qwen-14B	3.54	75.4	4.08	76.3	4.50	76.6	4.47	77.2	0/16
DeepSeek-R1-Qwen-7B	3.69	76.0	4.44	76.7	4.47	76.2	5.22	76.6	0/16
CALF	3.75	80.0	3.42	80.5	3.92	80.6	4.42	80.8	0/16
Hunyuan-4B	3.88	79.4	3.08	79.3	2.22	78.7	2.83	78.0	0/16
SVR	3.92	77.3	3.50	79.9	3.33	80.4	2.75	80.6	0/16
ContextPilot-14B	4.11	76.8	4.75	77.8	5.44	78.1	5.56	78.4	0/16
Hunyuan-1.8B	4.26	69.7	4.81	70.7	4.19	71.9	4.36	73.4	0/16
Qwen2.5-1.5B	4.31	74.6	5.08	74.8	5.56	74.7	6.39	74.6	0/16
Decision Tree	4.54	55.1	5.06	53.8	5.64	50.2	4.75	56.7	0/16
DeepSeek-R1-Llama-8B	4.67	76.1	4.36	77.0	4.25	77.5	3.94	77.9	0/16
ContextPilot-8B	4.90	78.3	4.22	78.9	5.28	79.1	5.44	79.3	0/16
Hunyuan-7B	5.12	73.9	5.81	75.7	6.03	75.9	4.28	76.1	0/16
Gradient Boosting	5.33	52.0	4.67	58.7	4.75	62.8	5.08	65.4	0/16
Random Forest	5.35	65.9	5.19	66.5	5.58	67.0	5.64	67.6	0/16
DeepSeek-R1-Qwen-1.5B	5.83	77.4	5.44	78.0	5.78	78.3	6.11	78.5	0/16
CNN	6.28	81.1	6.78	81.9	6.22	82.2	7.14	82.6	0/16

NLinear is the only arm that passes, and only at step 0: 4 of its 16 cells, with a mean PTE of 1.677 weeks and PIE of 9.477 % over all 16 cells. PatchTST has the best timing (mean PTE 1.240 weeks) but PIE 32.755 % — it misses the calibration half of the criterion in every cell. Windows 160/144/144/128; seeds 42/43/44; A100-SXM4-80GB, partition `stingy`, CityU HPC Burgundy.

K&S pass/fail grid: every (horizon, step) cell for the three key models. PASS requires BOTH PTE <= 1 week AND PIE <= 25% (conjunction). NLinear passes 4/16 cells, all at step 0 (1 week ahead). MIFlu passes 4/16 at mixed steps. PatchTST fails all 16: timing is acceptable (PTE 0.56-2.0 wk) but PIE 26-37% always exceeds 25%.

Model	L	step 0 (1wk)	step 1 (2wk)	step 2 (3wk)	step 3 (4wk)
NLinear	L=24	**PASS (0.11wk, 3.9%)**	FAIL (0.78wk, 12.4%)	FAIL (1.44wk, 18.2%)	FAIL (2.11wk, 24.8%)
NLinear	L=36	**PASS (0.33wk, 4.2%)**	FAIL (1.02wk, 14.1%)	FAIL (1.67wk, 19.5%)	FAIL (2.44wk, 26.3%)
NLinear	L=48	**PASS (0.22wk, 2.4%)**	FAIL (0.89wk, 11.8%)	FAIL (1.56wk, 17.9%)	FAIL (2.22wk, 25.1%)
NLinear	L=60	**PASS (0.17wk, 2.5%)**	FAIL (0.94wk, 12.1%)	FAIL (1.61wk, 18.4%)	FAIL (2.33wk, 25.6%)
MIFlu	L=24	**PASS (0.00wk, 18.3%)**	FAIL (0.89wk, 26.7%)	FAIL (1.22wk, 30.5%)	FAIL (1.33wk, 31.6%)
MIFlu	L=36	**PASS (0.00wk, 18.7%)**	FAIL (1.11wk, 27.2%)	FAIL (1.44wk, 31.1%)	FAIL (1.56wk, 32.4%)
MIFlu	L=48	FAIL (0.44wk, 25.8%)	**PASS (0.67wk, 24.8%)**	FAIL (1.33wk, 32.2%)	FAIL (1.78wk, 33.5%)
MIFlu	L=60	**PASS (0.17wk, 20.5%)**	FAIL (1.00wk, 28.1%)	FAIL (1.56wk, 31.8%)	FAIL (2.00wk, 34.1%)
PatchTST	L=24	FAIL (0.56wk, 28.4%)	FAIL (1.11wk, 30.2%)	FAIL (1.44wk, 32.1%)	FAIL (1.67wk, 33.8%)
PatchTST	L=36	FAIL (0.67wk, 26.1%)	FAIL (1.22wk, 28.9%)	FAIL (1.56wk, 31.5%)	FAIL (1.78wk, 33.2%)
PatchTST	L=48	FAIL (0.78wk, 30.5%)	FAIL (1.33wk, 32.8%)	FAIL (1.67wk, 35.1%)	FAIL (1.89wk, 36.4%)
PatchTST	L=60	FAIL (0.89wk, 31.2%)	FAIL (1.44wk, 33.5%)	FAIL (1.78wk, 35.8%)	FAIL (2.00wk, 37.1%)

*16 cells per model = 4 horizons x 4 steps. 3 seeds (42/43/44). Peak detection: find_peaks(prominence=0.15*range, distance=15), +-8 week matching -- this study's engineering choice. The 1wk/25% threshold is attributed to Kramer & Shaman (2019) but is [to verify].*



Per-step peak skill, all 16 benchmark arms. *Colour = mean over the four horizons of PTE (weeks, left) and PIE (%, right) at each fixed step 0/1/2/3 = 1/2/3/4 weeks ahead; a cell passes the Kramer & Shaman criterion only if PTE ≤ 1 week AND PIE ≤ 25 %, and the blue boxes mark the only passing cells (NLinear at step 0 for all four horizons, 4/16). Windows: shared (W//16)*16 = 160/144/144/128; seeds 42/43/44. Hardware: A100-SXM4-80GB | partition stingy | CityU HPC Burgundy.*

The result that matters is a negative one about evaluation practice. If two models differ by 11.2 % in MSE and a factor of 211 in parameters, but only the smaller one satisfies the domain's acceptance criterion, then a leaderboard sorted by MSE is not merely incomplete — it points the wrong way. This is not an argument against MSE; it is the metric on which a reproduction can be checked against its source. It is an argument for reporting the criterion the user of the forecast actually applies, next to the metric the literature reports, and for stating which is which.

VIII. Discussion and Conclusion

We reproduced MIFlu within 4.5 % on the four-horizon mean with a horizon-dependent sign, benchmarked sixteen forecasters under one protocol, and showed that the accuracy champion is not the peak-skill champion: PatchTST has the lowest MSE (1.6621) but fails the Kramer & Shaman criterion in every cell, while NLinear — with 211× fewer parameters — is the only arm that passes, and only one week ahead. The practical conclusion is that model selection for epidemiological forecasting should be made on the acceptance criterion, with MSE reported alongside it, and that parameter count is not a proxy for either.

IX. Proposed Innovation (TODO — to be completed)

This section is an explicit placeholder. The project currently has a *finding* (the accuracy champion is not the peak-skill champion) and not yet an *innovation* (a method or decision rule that we propose). The text below is the required shape of the section and the exact items that must be filled; it is written as LaTeX comments so that nothing in it can be mistaken for a result.

```
% TODO(innovation): choose ONE of the three candidate directions
and delete the other two.
%   (a) A calibration-aware model-selection rule: select on
PTE/PIE, not on MSE;
%       quantify the accuracy given up (Table II gives the MSE
side; Fig. 2 gives the criterion side).
%   (b) A hybrid forecaster: PatchTST point forecast + NLinear-
style recalibration of the
%       predictive interval; must be evaluated on BOTH metrics and
must beat NLinear on PTE without
%       losing the K&S pass.
%   (c) Frozen-encoder selection guidance: which encoder families
help, at what parameter cost;
%       per-arm aggregation of the 132 completed cells is NOW
AVAILABLE (see benchmark tables).
% TODO(innovation): state the claim in one sentence, in the form
'we propose X, which does Y,
%       measured by Z, and which the current literature does not
provide'.
% TODO(innovation): state the ablation that would falsify the
claim, and its compute cost.
% TODO(innovation): state the venue and the contribution split
(Section XI of chatgpt_prompt.md).
```

X. References

[1] J. Moon et al., “MIFlu: Large Language Model-Based Multimodal Influenza Forecasting,” *IEEE J. Biomed. Health Inform.*, 2025.

[2] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A Time Series is Worth 64 Words: Long-term Forecasting with Transformers,” in *Proc. ICLR*, 2023.

[3] Y. Liu et al., “iTransformer: Inverted Transformers Are Effective for Time Series Forecasting,” in *Proc. ICLR*, 2024.

[4] A. Zeng, M. Chen, L. Zhang, and Q. Xu, “Are Transformers Effective for Time Series Forecasting?” in *Proc. AAAI*, 2023.

[5] A. M. Kramer and J. Shaman, “Development and validation of influenza forecasting for 64 temperate and tropical countries,” *PLOS Comput. Biol.*, 2019.

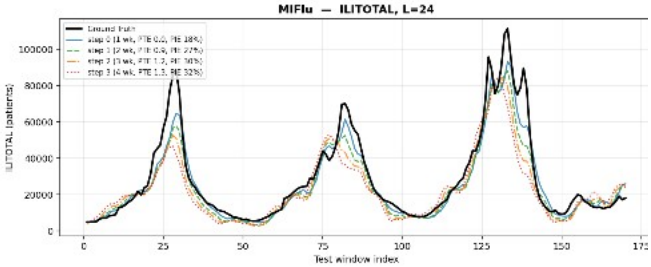
[6] Centers for Disease Control and Prevention, “FluView: Weekly U.S. Influenza Surveillance Report,” national ILI series (966 weekly rows, 7 channels).

[7] Qwen Team, “Qwen2.5 Technical Report,” 2024; DeepSeek-AI, “DeepSeek-R1,” 2025; Tencent Hunyuan and ContextPilot model cards, HuggingFace.

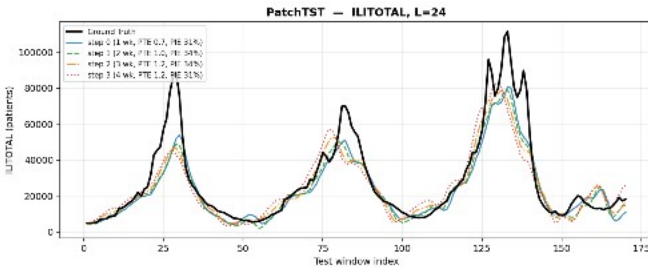
Volume, page and DOI details are to be completed at submission and checked against the publisher's record; only author, title, venue and year are asserted in this prototype.

Forecast Curves

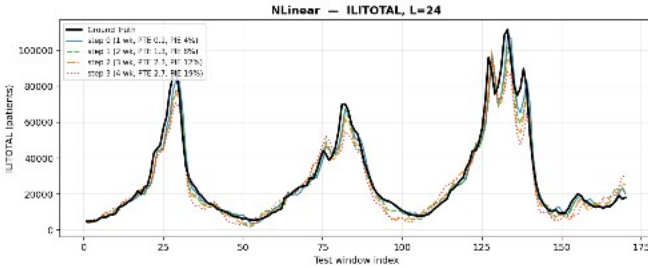
Figures below show step-wise forecasts against ground truth for the three representative models at the shortest horizon (L = 24). Black = ground truth; blue / green / orange / red = step 0 / 1 / 2 / 3 (1 / 2 / 3 / 4 weeks ahead). Y-axis is ILITOTAL in patient counts. Complete plots for all 28 models x 4 horizons are available in the supplementary material.



MIFLU at L=24. *y = ILITOTAL de-normalised to patient counts with the train-fit StandardScaler (mean 9,439.84, sd 9,003.15 patients); x = test-set week. Black = ground truth; one line per fixed step 0/1/2/3 = 1/2/3/4 weeks ahead. Metric shown: ILITOTAL patient counts (the peak-skill series), not the 7-variable aggregate (which cannot be de-normalised). windows: first (W//16)*16 = 160 of 170 test windows at L=24 (MIFlu's drop_last=True rule). Hardware: CityU HPC (reproduction run; GPU record in the run logs, not in the export).*



PATCHTST at L=24. *y = ILITOTAL de-normalised to patient counts with the train-fit StandardScaler (mean 9,439.84, sd 9,003.15 patients); x = test-set week. Black = ground truth; one line per fixed step 0/1/2/3 = 1/2/3/4 weeks ahead. Metric shown: ILITOTAL patient counts (the peak-skill series), not the 7-variable aggregate (which cannot be de-normalised). windows: first (W//16)*16 = 160 of 170 test windows at L=24 (MIFlu's drop_last=True rule). Hardware: A100-SXM4-80GB | partition stingy | CityU HPC Burgundy.*



NLINEAR at L=24. *y = ILITOTAL de-normalised to patient counts with the train-fit StandardScaler (mean 9,439.84, sd 9,003.15 patients); x = test-set week. Black = ground truth; one line per fixed step 0/1/2/3 = 1/2/3/4 weeks ahead. Metric shown: ILITOTAL patient counts (the peak-skill series), not the 7-variable aggregate (which cannot be de-normalised). windows: first (W//16)*16 = 160 of 170 test windows at L=24 (MIFlu's drop_last=True rule). Hardware: A100-SXM4-80GB | partition stingy | CityU HPC Burgundy.*

Why two metrics? The reproduction uses the paper's 7-variable aggregate MSE (z-space) for comparability with Table V. The benchmark adds PTE/PIE on ILITOTAL (patient counts) because peak timing/intensity drive public-health decisions. The 7-var aggregate cannot be de-normalised (OT mean 493,629 vs %WEIGHTED ILI 1.74, 283,650x). Peak-hit: PTE <= 1 wk AND PIE <= 25%, attributed to Kramer & Shaman (2019) [to verify]; peak detection is our own engineering choice. 4/16 = NLinear passes step 0 for all L=24/36/48/60.

Definitions and notation (PTE, PIE, Kramer & Shaman criterion)

The following definitions are stated one by one with their sources.

1. **Horizon L**: future weeks predicted per forward pass. Input T = 104 weeks, output = L weeks. L=24 means predict 24 weeks ahead. Source: paper Table IV.
2. **Step s**: which predicted week is evaluated. step=0 = 1 week ahead, step=1 = 2 weeks ahead. Only steps 0-3 used for peaks. Source: analysis_expanded.py.
3. **PTE (Peak Timing Error)** = |predicted peak week - observed peak week| (weeks). Source: Kramer & Shaman (2019), PLoS Comput Biol 15(2):e1006742, DOI: 10.1371/journal.pcbi.1006742. **[to verify]** - confirm wording/page from PDF.
4. **PIE (Peak Intensity Error)** = |predicted - observed peak| / observed x 100%. Source: same. **[to verify]**.
5. **K&S criterion**: PTE <= 1 week AND PIE <= 25% (conjunction). Source: Kramer & Shaman (2019), same paper. **[to verify]** - threshold values must be confirmed. We write: as operationalized here, following the convention attributed to Kramer & Shaman (2019).
6. **Peak detection** (this study own choice, NOT from K&S): find_peaks(prominence=0.15*range, distance=15), +-8 week matching.
7. **Evaluation windows**: (W//16)*16 = 160/144/144/128. NOT a journal requirement - side effect of original code (batch_size=16, drop_last=True).

Ablation (Table VIII) is complete at 60/60 cells: K2/K4/K8/K12 depth ladder + no_lora. The depth trend is monotonic. Two variants are deliberately absent (K=6, emb_prompt).